



Figure 1: Creating the project

APIs, which enable the apps to use a device's native features.

The framework includes:

- Accelerometer
- Camera
- Geolocation
- Storage
- Media
- Notification
- SplashScreen
- Contacts and many more features

My system's configuration:

- Linux distro: Ubuntu 12.04 32-bit desktop version
- Eclipse: Juno
- Android version: 4.1
- PhoneGap: Cordova 2.7.0

All the following results and screenshots are with respect to the above configurations.

Setting up the environment

To get started with the actual development, set up an environment where some installations and configurations need to be done.

The minimum requirements are:

1. Download and install an IDE – Eclipse is recommended
2. Download the Android SDK and the ADT plug-in (Android 2.2, 2.3 and 4.x are supported)
3. Download and extract the latest PhoneGap package – Cordova 2.7.0 (this is the latest one)



Note: The Android_SDK comes in different versions for different operating systems—*android_sdk_windows*, *android_sdk_linux* and *android_sdk_mac*

The project set-up

Pre-requisites: All the tools need to be downloaded and installed.

- You need an IDE installed to provide the development environment. After it gets installed, open its workspace and choose a workspace.

- Set up the Android environment in Eclipse by installing the ADT plug-in there.
- Set up the path for *android_sdk* by going to *Windows > Preferences > Android > Browse*; this is the path to the SDK folder.

That's it—we have the environment set for the development now. Let's dive into it.

After setting up everything, start integration with PhoneGap.

- Create a new project by pointing your cursor to *File > New > Android Project*.
- Specify the project name (Let's call it: PhoneGapDemo). The same is depicted in Figure 1.
- Specify the 'Build Target'. Any target can be chosen, i.e., 2.2, 3.2, 4.0, etc. (Let's choose 2.3).
- Specify the 'Application Name' (Let's call it MyPhoneGap).
- Specify the 'Package Name'. The package name must start with 'com' (com.phonegap).
- Specify the 'Activity Name'. It is a Java file name (MainActivity.java).
- Click 'Finish'. We are done with building a new project and it will be on your Project Explorer now. The screen shot in Figure 2 depicts this.

Now, let's integrate this project with the PhoneGap we downloaded earlier, as follows:

- Create a new directory (folder) and name it *libs*.
- Open your PhoneGap package and go to the folder *Android* and then copy the *.jar* file found there. Paste this file into the *libs* folder that you just created.
- Create a folder under the assets folder and name it *www*. Copy the *.js* file from the same PhoneGap package, *Phonegap>Android* and place it in *www*.
- Set the build path. Right click on *Project> Build Path> Configure Build Path> Libraries> Add Jar > Add Phonegap.jar* file.
- Paste all the *.css* files in *www* if you already have some
- Create an *xml* folder inside the *res* directory.
- Copy the *config.xml* file, which is in the PhoneGap package that you downloaded—*phonegap-2.7.0/lib/android/xml/config.xml*—and paste it into the *xml* folder.



Note: *config.xml* contains all the plugins needed for using PhoneGap APIs.

The boring integration part is over, so let's start the real development.

Developing 'Hello world'

The steps for the development are as follows:

1. Create an HTML page inside the *www* folder.
2. Modify the Java class *MyPhoneGapActivity.java*.
3. Add some permissions in the *AndroidManifest.xml*.
4. Deploy and run the application on the emulator.